CSCI 49376 Assignment 1

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Big Data Technology · Prof. Alex Liew · Parts 1–4

Part 1. Select and Examine the Original Dataset

#	Required	Answer
1	Dataset selected	Option 2: Diabetes 130-US Hospitals for Years 1999–2008 (Kaggle: brandao/diabetes)
2	CSV file used	`diabetic_data.csv`
3	Total rows	101,766 (one row per hospital stay)
4	Total columns	50 (24 ignored as instructed, 26 used)

5–6. Attributes and what each represents

Attribute	Description
encounter_id	Unique ID of one hospital stay
patient_nbr	ID of a patient (a patient can have several stays)
race	Caucasian, AfricanAmerican, Hispanic, Asian, Other
gender	Female, Male, Unknown/Invalid
age	Age group in 10-year bands, e.g. [70-80)
weight	Weight band in pounds
admission_type_id	Code: type of admission (e.g. emergency, elective)
discharge_disposition_id	Code: where the patient went at discharge (e.g. home)
admission_source_id	Code: where the patient came from (e.g. referral, ER)
time_in_hospital	Days in the hospital (1–14)
payer_code	Insurance code (e.g. MC = Medicare)
medical_specialty	Specialty of the admitting doctor
num_lab_procedures	Lab tests during the stay
num_procedures	Other procedures during the stay
num_medications	Distinct medications given
number_outpatient	Outpatient visits in the prior year
number_emergency	Emergency visits in the prior year
number_inpatient	Inpatient visits in the prior year
diag_1	Primary diagnosis (ICD-9 code)
diag_2	Secondary diagnosis (ICD-9 code)
diag_3	Additional diagnosis (ICD-9 code)
number_diagnoses	Number of diagnoses recorded
max_glu_serum	Glucose test: >200, >300, Norm, None (not measured)
A1Cresult	HbA1c test: >7, >8, Norm, None (not measured)
diabetesMed	Any diabetes medication prescribed: Yes/No
readmitted	NO, >30 (after 30 days), <30 (within 30 days)

Attribute	Description
metformin, repaglinide, nateglinide, chlorpropamide, glimepiride, acetohexamide, glipizide, glyburide, tolbutamide, pioglitazone, rosiglitazone, acarbose, miglitol, troglitazone, tolazamide, examide, citoglipton, insulin, glyburide-metformin, glipizide-metformin, glimepiride-pioglitazone, metformin-rosiglitazone, metformin-pioglitazone, change	Ignored, as instructed: 23 medication columns (No, Steady, Up, Down) and whether any medication changed

7. Possible identifiers or keys

Column	Finding
encounter_id	Unique on all 101,766 rows → primary key of the original table
patient_nbr	71,518 distinct values; repeats across stays → identifies a separate PATIENT entity
admission_type_id, discharge_disposition_id, admission_source_id	8, 26 and 17 codes → reference tables (descriptions in `IDs_mapping.csv`, UCI)
diag_1, diag_2, diag_3	915 distinct ICD-9 codes → a DIAGNOSIS entity

8. Missing, unknown or incomplete values

Column	Missing	Written as
weight	96.9%	?
max_glu_serum	94.7%	None (not measured)
A1Cresult	83.3%	None (not measured)
medical_specialty	49.1%	?
payer_code	39.6%	?
race	2.2%	?
diag_3 / diag_2 / diag_1	1.4% / 0.4% / 21 rows	?
gender	3 rows	Unknown/Invalid

All are stored as NULL. Inconsistencies: 123 patients have two different race values and 3 have two genders across stays (we keep the value from the patient's latest stay); age changes between stays for 1,574 patients, so age belongs to the stay.

Part 2. Normalize the Dataset

Original relation → 1NF → 2NF. Rows 1–6 are the six items required for each stage.

	Original relation	1NF	2NF (final design)
1. Structure	DIABETIC_DATA(encounter_id, patient_nbr, race, gender, age, …, diag_1, diag_2, diag_3, …, readmitted): one table, one row per stay	ENCOUNTER_1NF(encounter_id, diag_position, icd9_code, patient_nbr, race, gender, age, …, readmitted): one row per diagnosis	7 tables, listed below
2. Keys	encounter_id	(encounter_id, diag_position)	One per table, listed below
3. Functional dependencies	encounter_id → every other column patient_nbr → race, gender each code → its description	FD1: (encounter_id, diag_position) → icd9_code FD2: encounter_id → age, time_in_hospital, …, readmitted FD3: patient_nbr → race, gender FD4: each code → description	FD1 → ENCOUNTER_DIAGNOSIS FD2 → ENCOUNTER FD3 → PATIENT FD4 → the three code tables
4. Redundancy or dependency problems	diag_1–diag_3 are a repeating group (three columns for one kind of value). Patient data repeats on every stay (30,248 extra copies).	Partial dependency (FD2): stay columns depend on only part of the key, so each stay's details are copied once per diagnosis.	None: no partial dependencies, and each fact is stored once.

	Original relation	1NF	2NF (final design)
5. Changes to reach the next form	Turn the three diagnosis columns into one row per diagnosis and add diag_position (1, 2, 3).	Move FD2's columns to ENCOUNTER, keyed by encounter_id. To minimize redundancy, also move FD3 to PATIENT and FD4 to code tables.	Final design.
6. Why it satisfies the form	Not in 1NF (repeating group).	Every cell holds one value, and the repeating group is gone.	Every non-key column depends on the whole key: one-column keys can't have partial dependencies, and in ENCOUNTER_DIAGNOSIS icd9_code needs both key columns.

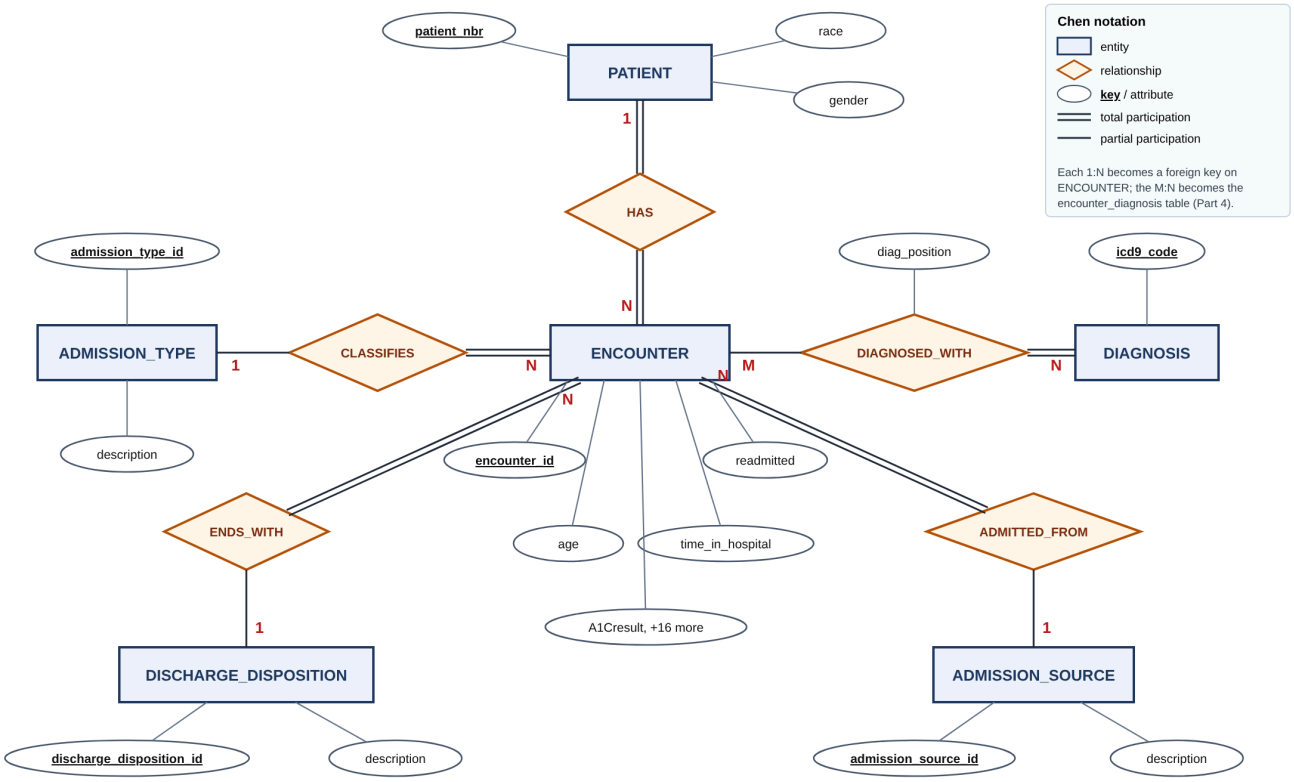
The key uses diag_position, not icd9_code, because 5,492 stays list the same code twice.

Final tables

Table	Attributes	Primary key	Foreign key(s)
PATIENT	patient_nbr, race, gender	patient_nbr	—
ENCOUNTER	encounter_id, patient_nbr, admission_type_id, discharge_disposition_id, admission_source_id, age, weight, time_in_hospital, payer_code, medical_specialty, num_lab_procedures, num_procedures, num_medications, number_outpatient, number_emergency, number_inpatient, number_diagnoses, max_glu_serum, A1Cresult, diabetesMed, readmitted	encounter_id	patient_nbr → PATIENT admission_type_id → ADMISSION_TYPE discharge_disposition_id → DISCHARGE_DISPOSITION admission_source_id → ADMISSION_SOURCE
ENCOUNTER_DIAGNOSIS	encounter_id, diag_position, icd9_code	(encounter_id, diag_position)	encounter_id → ENCOUNTER icd9_code → DIAGNOSIS
DIAGNOSIS	icd9_code	icd9_code	—
ADMISSION_TYPE	admission_type_id, description	admission_type_id	—
DISCHARGE_DISPOSITION	discharge_disposition_id, description	discharge_disposition_id	—
ADMISSION_SOURCE	admission_source_id, description	admission_source_id	—

Nothing is lost: joining the tables rebuilds every original row. Expected rows: ENCOUNTER 101,766 · PATIENT 71,518 · ENCOUNTER_DIAGNOSIS 303,496 · DIAGNOSIS 915 · code tables 8 / 26 / 17.

Part 3. Construct an ER Diagram



Entities

Entity	Important attributes	Primary key	Foreign keys
PATIENT	race, gender	patient_nbr	—
ENCOUNTER	age, time_in_hospital, A1Cresult, readmitted	encounter_id	patient_nbr, admission_type_id, discharge_disposition_id, admission_source_id
DIAGNOSIS	—	icd9_code	—
ADMISSION_TYPE	description	admission_type_id	—
DISCHARGE_DISPOSITION	description	discharge_disposition_id	—
ADMISSION_SOURCE	description	admission_source_id	—

Relationships, cardinalities and participation

Relationship	Between	Type	Why	Participation
has	PATIENT – ENCOUNTER	1:N	A patient has many stays; a stay has one patient.	Total on both sides
classifies	ADMISSION_TYPE – ENCOUNTER	1:N	A type covers many stays; a stay has one type.	Total for ENCOUNTER, partial for the type
ends_with	DISCHARGE_DISPOSITION – ENCOUNTER	1:N	Same pattern.	Total for ENCOUNTER, partial for the disposition
admitted_from	ADMISSION_SOURCE – ENCOUNTER	1:N	Same pattern.	Total for ENCOUNTER, partial for the source
diagnosed_with	ENCOUNTER – DIAGNOSIS	M:N	A stay has up to 3 diagnoses; a diagnosis appears in many stays. Attribute: diag_position.	Partial for ENCOUNTER (some stays have none), total for DIAGNOSIS

There are no 1:1 relationships. Notation: rectangle = entity, diamond = relationship, oval = attribute (key underlined), double line = total participation.

Part 4. Convert the ER Diagram into Relational Tables

Relation	Attributes	Primary key	Foreign key(s)
PATIENT	patient_nbr, race, gender	patient_nbr	—
ADMISSION_TYPE	admission_type_id, description	admission_type_id	—
DISCHARGE_DISPOSITION	discharge_disposition_id, description	discharge_disposition_id	—
ADMISSION_SOURCE	admission_source_id, description	admission_source_id	—
DIAGNOSIS	icd9_code	icd9_code	—
ENCOUNTER	encounter_id, patient_nbr, admission_type_id, discharge_disposition_id, admission_source_id, age, weight, time_in_hospital, payer_code, medical_specialty, num_lab_procedures, num_procedures, num_medications, number_outpatient, number_emergency, number_inpatient, number_diagnoses, max_glu_serum, A1Cresult, diabetesMed, readmitted	encounter_id	patient_nbr, admission_type_id, discharge_disposition_id, admission_source_id
ENCOUNTER_DIAGNOSIS	encounter_id, diag_position, icd9_code	(encounter_id, diag_position)	encounter_id, icd9_code

How the ER relationships are represented

ER relationship	Type	Represented as
PATIENT has ENCOUNTER	1:N	Foreign key ENCOUNTER.patient_nbr → PATIENT (on the N side, NOT NULL)
ADMISSION_TYPE classifies ENCOUNTER	1:N	Foreign key ENCOUNTER.admission_type_id → ADMISSION_TYPE
DISCHARGE_DISPOSITION ends_with ENCOUNTER	1:N	Foreign key ENCOUNTER.discharge_disposition_id → DISCHARGE_DISPOSITION
ADMISSION_SOURCE admitted_from ENCOUNTER	1:N	Foreign key ENCOUNTER.admission_source_id → ADMISSION_SOURCE
ENCOUNTER diagnosed_with DIAGNOSIS	M:N	Its own table, ENCOUNTER_DIAGNOSIS: a foreign key to each side plus the relationship attribute diag_position

Table creation order: PATIENT, ADMISSION_TYPE, DISCHARGE_DISPOSITION, ADMISSION_SOURCE, DIAGNOSIS → ENCOUNTER → ENCOUNTER_DIAGNOSIS.

Table diagram

The same schema drawn as tables, with planned MySQL types (main ENCOUNTER columns only).

